TEAM 050 · FIRST DRAFT FOR DISCUSSION

Topic tags for mentor matching

A first list built from our interviews and the 15 October session notes. We would like IIC to correct it.

Matching works by **topic tags**. Students tag the problem they need help with, IIC tags each mentor with the areas they have real experience in, and the platform recommends mentors who share the most tags. Recommendations can only be as good as this list, so we would like IIC to check it before we build it into the system.

Students
Pick 1–5 tags when they submit a problem, and describe it in their own words.

Mentors
Tagged by IIC when added, only for areas of real experience. We suggest at most 8 tags per mentor.

Recommendations
Up to 4 available mentors, ranked by shared tags, then by fewest current mentees. IIC makes the final choice.

Gaps
Tags that students choose but no mentor has are reported, so IIC knows whom to invite next cycle.

Why the tags are narrow. One student told us they wanted "not broad terms like data science … maybe a mentor for computer vision". A broad tag would match almost every mentor and tell IIC nothing. The list below therefore names specific skills, decisions and situations.

The draft list: 28 tags in 7 groups

S01 S02 student interviews · M01 alumnus interview · IIC notes from the 15 October session ("specific industry requirements, how to improve DSA & SQL skills") · PS project problem statement · Proposed not raised in our research; added to cover an obvious gap, so please keep or drop · Confirm scope close to placement, which is out of scope

#	Tag	What it covers	Source
Career direction			
1	**Choosing a career path**	Deciding between a job, higher studies, a startup or a family business, and which suits the student's situation	S01 S02 PS
2	**Industry and role requirements**	What a specific industry or role expects from graduates, and how to close the gap	IIC PS
Jobs and internships (career guidance, not the placement process)			
3	**Finding internships**	Where and how to look for internships, including research internships	M01 S01
4	**Job search strategy**	Which roles to target and how to apply beyond campus channels	M01
5	**Technical interview preparation**	Coding, SQL and database-design rounds	M01 Confirm scope
6	**Resume and portfolio**	Presenting skills and projects to employers	Proposed
Technical skills			
7	**Data structures and algorithms (DSA)**	Improving DSA for interviews and projects	IIC
8	**SQL and database design**	Querying, schema design, database rounds	IIC M01

#	Tag	What it covers	Source
9	**Python programming**	Writing and structuring Python beyond coursework	M01
10	**Deep learning**	Models, training and applying deep learning	S01
11	**Computer vision**	Image tasks such as detection and segmentation	S01
12	**Speech technology**	Speech recognition and processing	S01
13	**Natural language processing**	Text-based machine learning	Proposed
14	**Data and business analytics**	Analysis, dashboards and analyst roles	Proposed
15	**Web and app development**	Building and deploying applications	Proposed
Higher studies and research			
16	**Master's or PhD abroad**	Choosing programmes and preparing applications	S01
17	**Research careers**	Getting into research, research internships, what a PhD involves	S01
18	**GATE and higher studies in India**	GATE preparation and what to focus on	M01
19	**Computational neuroscience**	An example of a niche research area; others would be added the same way	S01
Courses and academics			
20	**Choosing electives**	Especially rarely taken electives, where few seniors can be found	S01 S02
21	**Planning a term's course load**	How many and which courses to take, given time and capacity	M01 S02
22	**Getting through a difficult course**	Approach, resources and notes for a hard course	S01 S02
23	**Improving CGPA**	Planning to raise grades over the coming terms	S02
Projects			
24	**Stuck on a course project**	Direction when a project is blocked; mentors guide rather than do the work	M01
25	**Niche or research project**	Finding someone who has worked on a similar problem	S01
Entrepreneurship			
26	**Starting a startup**	Testing an idea and taking the first steps	S02 PS
27	**Joining or growing a family business**	Bringing degree skills into an existing business	S02
Catch-all			
28	**Other**	The student describes the need; IIC reviews it. Needs that keep recurring become new tags next cycle.	Design

How a problem gets matched: an example

The problem below is paraphrased and anonymised from one of our student interviews. The mentors are made up.

Student's problem

"I'm considering the speech technology and deep learning for computer vision electives next term, but I can't find anyone who has taken them. I'd like to hear from someone who has worked in these areas before I decide."

Tags chosen:

20 Choosing electives · 11 Computer vision · 12 Speech technology

What IIC sees

Mentor	Shared tags	Mentees
Mentor A: CV engineer	**2**	0 of 1
Mentor B: speech research	**1**	0 of 1
Mentor C: speech + NLP	**1**	1 of 2
Mentor D: data analyst	**0**	not shown

A ranks first with the most shared tags. B ranks above C because, with the same number of shared tags, B has fewer current mentees. IIC reads the problem and picks one of them.

Not handled through tags

Need	What we propose
Feeling overwhelmed, or other wellbeing concerns	Not matched to a mentor. The request reaches IIC, which refers the student to the right support. Where should these go?
Placement portal and placement process	Out of scope, as agreed on 15 October. Tags 3–5 cover career guidance about jobs, not the placement process itself.

Questions for IIC

Once IIC is happy with the list, we will load it as fixed seed data. Students and mentors will choose from it, and it will be used when importing mentors.

1	What do students most often ask IIC about that is missing from this list?
2	Should tags 3–5 (internships, job search, interview preparation) stay, given that placement is out of scope? Where is the line?
3	Should we keep or drop the four proposed tags: 6 Resume and portfolio, 13 NLP, 14 Data and business analytics, 15 Web and app development?
4	Should elective guidance be tagged by subject area, as here, or by named course from the course catalogue?
5	Should IIC be able to add, rename or retire tags each cycle? This would add one small feature to our plan.
6	For the dummy mentor database, what backgrounds do mentors typically have, so that we can tag them realistically?